

```

# Install required libraries
!pip install opencv-python matplotlib scikit-image

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from skimage.filters import sobel

# -----
# Upload Image
# -----
uploaded = files.upload()
image_path = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)

# -----
# SOBEL OPERATOR
# -----
sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_mag = np.sqrt(sobelx**2 + sobely**2)
sobel_dir = np.arctan2(sobely, sobelx)

# -----
# PREWITT OPERATOR
# -----
# Define kernels manually
kernelx = np.array([[ -1,0,1],
                    [ -1,0,1],
                    [ -1,0,1]])

kernely = np.array([[ -1,-1,-1],
                    [ 0,0,0],
                    [ 1,1,1]])

prewittx = cv2.filter2D(gray, -1, kernelx)
prewitty = cv2.filter2D(gray, -1, kernely)

prewitt_mag = np.sqrt(prewittx**2 + prewitty**2)
prewitt_dir = np.arctan2(prewitty, prewittx)

# -----
# Sample Gradient Values (Center Pixel)
# -----
h, w = gray.shape
cx, cy = h//2, w//2

print("----- SAMPLE GRADIENT VALUES -----")
print("SOBEL -> Gx:", sobelx[cx,cy], "Gy:", sobely[cx,cy], "Magnitude:", sobel_mag[cx,cy])
print("PREWITT -> Gx:", prewittx[cx,cy], "Gy:", prewitty[cx,cy], "Magnitude:", prewitt_mag[cx,cy])

# -----
# Normalize for display
# -----
sobel_disp = cv2.normalize(sobel_mag, None, 0, 255, cv2.NORM_MINMAX)
prewitt_disp = cv2.normalize(prewitt_mag, None, 0, 255, cv2.NORM_MINMAX)

# -----
# Display Results
# -----
titles = [
    "Original Image", "Grayscale",
    "Sobel Edge Magnitude", "Prewitt Edge Magnitude",
    "Sobel Direction", "Prewitt Direction"
]

images = [
    img, gray,
    sobel_disp, prewitt_disp,
    sobel_dir, prewitt_dir
]

plt.figure(figsize=(12, 8))

```

```

plt.figure(figsize=(10,10))
for i in range(len(images)):
    plt.subplot(2,3,i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis('off')

plt.tight_layout()
plt.show()

```

Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
 Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
 Requirement already satisfied: scikit-image in /usr/local/lib/python3.12/dist-packages (0.25.2)
 Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
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 Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (2026.4.11)
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 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

[Choose Files](#) KGE_5TKp...85GBjqb.jpg

KGE_5TKpQB8dRmCoZqk3UCIKMvMukqQNJeAgTP-O4FkXm3-

WYWDqUOXh4QnflRBg36zdUlxzyWVANW4AxElHmF3UbXRpu8SbZDC2prwOL0JepBtHSNNmsS2Oh9wnfhY04BUKzy1DXsQmAt2H6ul8YsNxlurCzAUguV8eSx8U85GBjqb.jpg (image/jpeg) - 40963 bytes, last modified: 7/22/2026 - 100% done

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----- SAMPLE GRADIENT VALUES -----

SOBEL -> Gx: 25.0 Gy: -21.0 Magnitude: 32.64965543462902

PREWITT -> Gx: 16 Gy: 0 Magnitude: 0.0

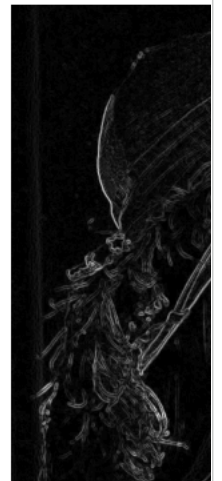
Original Image



Grayscale



Sobel E



Prewitt Edge Magnitude



Sobel Direction



Prew

